Molecular Evolution of Vocal Production Learning in Bats: A Review

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M.A. Thesis

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Spring 2022

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Abstract

Some, but not all, bats are vocal production learners. However, the evolutionary forces shaping vocal learning in bats is not well understood. Echolocation may prime bats for vocal learning by providing much of the necessary physiological machinery, while their complex social structures create the need for communication. Several key genes are implicated in the vocal learning pathway. Examining the evolutionary patterns of these genes in association with bat behavioral phenotypes will unravel the evolutionary processes shaping vocal production learning.

Vocal Production Learning in Bats

Quantifying Vocal Learning

Bats are one of only a few clades of mammals for whom we have good evidence of vocal production learning, the ability to modify vocal sounds in imitation of conspecifics or the environment. While, for example, songbirds such as mockingbirds may imitate a car alarm, bottlenose dolphins may give each other individual names. Given that vocal learning is a key component of human language, studying the genetic basis of vocal learning in other species and the processes shaping its evolution is of interdisciplinary interest (Jarvis, 2019).

Vocal learning is a polyphyletic, multidimensional trait that is difficult to quantify and best described by a continuum (Table 1). As it is often defined, all vocal learning animals compare vocal output to the memory of auditory input, but across species other dimensions vary, such as whether an individual learns new sounds (learned acquisition) or modifies existing sounds (social modification). For example, African gray parrots are exceptional vocal learners, capable of mimicking new sounds from a variety of species quickly at any time in their life, remembering these sounds for a long time, and reproducing them accurately. A less sophisticated type of vocal learning is call convergence, in which individuals learn to match their own call to the group call to indicate membership. While some bats have clearly demonstrated the ability to learn converging calls, so far as we currently know, most bats are not accurate imitators and do not invent new sounds, but rather modify existing ones. Bats are only known to imitate sounds from conspecifics. Furthermore, no studies have yet identified how long bats retain learned auditory information (Vernes et al., 2021).

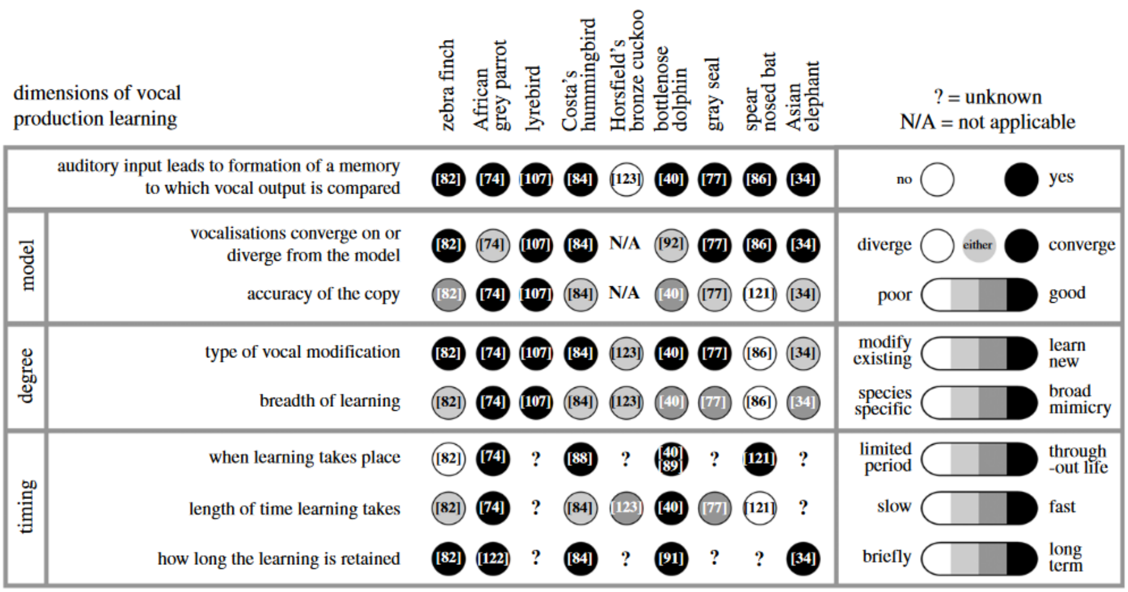

Table 1. Vocal learning can be broken down into a multi-dimensional model to compare the nature of vocal learning across species. From Vernes et al., 2021.

Evidence for vocal learning in mammals

Besides bats, other mammalian vocal learners include cetaceans, pinnipeds, elephants, and primates. Toothed whales, including bottlenose dolphins (*Tursiops truncatus*), Risso's dolphin (*Grampus griseus*), killer whales (*Orcinus orca*), and belugas (*Delphinapterus leucas*), have all been shown to be capable of learning new call types from either conspecifics or other species, such as humans. Groups of long-lived baleen whales, including bowhead whales (*Balaena mysticetus*) and humpback whales (*Megaptera novaeangliae*), have been observed to alter their songs over their lifetimes at a global population level, possible evidence of cultural transmission (Zandberg et al., 2021;Tervo et al., 2011;Mercado, 2021). Pinnipeds such as harbor seals (*Phoca vitulina*), gray seals (*Halichoerus grypus*), and walrus (*Odobenus rosmarus*) can imitate human vowel sounds in captivity, and wild groups may have dialects, although the evidence for that is less clear (Reichmuth & Casey, 2014). African elephants (*Loxodonta africana*) kept with Asian elephants (*Elephas maximus*) were found to imitate the sounds of the Asian elephants, and Asian elephants have been observed to copy human speech sounds, although no evidence exists of wild elephants using vocal modification in this manner (Stoeger & Manger, 2014). Surprisingly, vocal learning in non-human primates seems quite limited. Captive primates, including western lowland gorilla (*Gorilla gorilla*) and orangutans (*Pongo spp*.) have been shown capable of creating and imitating novel sounds, although very few of these are laryngeal. Other primates, such as gray mouse lemurs (*Microcebus murinus*), Campbell's monkeys (*Cercopithecus campbelli*), Guinea baboons (*Papio papio*), and chimpanzees (*Pan troglodytes*), demonstrated call convergence when exposed for long periods of time to other species, but no evidence yet exists for their ability to produce novel sounds (Fischer & Hammerschmidt, 2020; Janik & Knörnschild, 2021;Nowicki & Searcy, 2014). Given the few mammalian lineages capable of vocal learning, bats offer a unique opportunity to study the evolution of this trait.

Bat Social Structures Require Communication

Bats exhibit a remarkable diversity of social structure. Tropical bats roost together all year, while bats in temperate zones sometimes only socialize during certain seasons. Most species of bats form maternity colonies for communal raising of young, while males may be solitary, form male-only groups, or cohabit female colonies. The stability of these colonies varies across species, with some species following more complex fission-fusion dynamics. The degree of relatedness and size of groups also vary across species, with some groups averaging under ten individuals, while others may form colonies of thousands or millions (Kerth, 2008).

When living in groups, bats must frequently communicate to make locational and temporal decisions about roosting, foraging and mating. Logically, it follows that the degree or type of sociality likely relates to the strength of evolutionary pressure on vocal production learning in bats, but no studies have yet tested this. Bats have adapted both nonverbal and verbal communication. Verbal communication may be sonic or ultrasonic and shows a high diversity of frequency and type within and between species. The similarity between the social structures of bats and humans, who also form fission-fusion groups, makes bats a very interesting model for studying the evolution of vocal production learning (Vernes, 2017).

Direct Evidence for Vocal Learning in Bats

Bats have been suggested as a good mammalian system in which to study the evolution of vocal learning because of their sociability, their diversity, their gregariousness, and the ease of using some species in controlled lab settings (Knörnschild, 2014). Like parrots and dolphins, some bats use learned vocalizations to refer to individuals, while others likely use vocal learning for group recognition. Vocal learning has been directly observed in several species across the bat phylogeny (Figure

1). The greater horseshoe bat, *Rhinolophus ferrumequinum*, and the lesser spear-nosed bat, *Phyllostomus discolor*, both use vocal learning to help parents and offspring recognize one another (Lattenkamp et al., 2021; Esser, 1994). In one study, researchers were able to train adult *Phyllostomus discolor* to change the frequency and duration of their calls, direct evidence of vocal modification in response to auditory input memory (Lattenkamp & Vernes, 2018). Adult females of the greater spear-nosed bat, *Phyllostomus hastatus*, modify a foraging coordination call to converge to a group-specific frequency when group membership changes (Boughman, 1998). Captive Taiwanese leaf-nosed bats, *Hipposideros terasensis*, also adjust their frequency to match the colony-specific call, and geographic call variation in wild populations results from mother-pup cultural transmission, rather than genetic drift (Hiryu et al., 2006;Yoshino et al., 2008). The pups of Egyptian fruit bats, *Rousettus aegyptiacus*, take longer to learn the full range of their adult calls when raised with only their mother rather than an entire colony and have been shown to shift their call frequencies towards artificially raised or lowered calls (Prat et al., 2017), providing evidence of intertwined social and vocal learning. The pups of greater sac-winged bats, *Saccopteryx bilineata*, converge their calls to the group signature upon maturity (Knörnschild et al., 2012). Additionally, young bats of this species have been observed to go through a babbling-like practice phase of sound making, a behavior associated in other vertebrates, including humans, with vocal learning (Knörnschild et al., 2006).

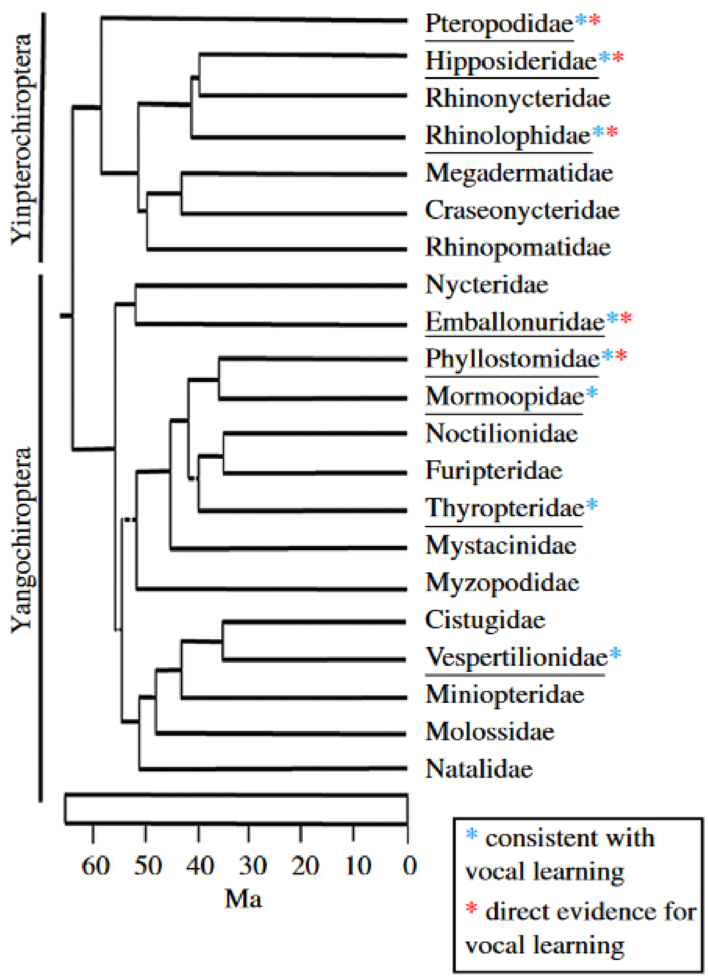

Figure 1. Evidence for vocal learning in bats mapped to the bat phylogenetic tree. Direct evidence was found in five families based on behavioral studies that found call convergence in adults or juveniles, juvenile call shift and juvenile-adult similarity. Evidence consistent with, but not exclusively indicative of vocal learning exists for eight families based on behavioral studies that found adult call similarity, geographical variation or group differences. From Vernes & Wilkinson, 2020.

Geographic Isolation as Evidence Consistent with Vocal Learning in Bats

Early studies in vocal learning relied heavily on differences in calls between geographically isolated populations. However, while this class of evidence is consistent with vocal learning, it is not exclusively indicative of vocal learning. For example, the Northern elephant seal (*Mirounga angustirostris*) was implicated early on as a vocal learner due to observed dialects in different populations, but a population bottleneck and founder effect turned out to be responsible (Casey et al., 2018). Similarly, geographical variation in the resting frequency (RF) of constant frequency (CF) emitting bats commonly has morphological underpinnings. Without direct evidence, we cannot tell if geographically isolated groups of bats make different calls because they are more genetically related or because younger bats are learning the call from older bats (Janik & Knörnschild, 2021). However, because geographically isolated dialects are consistent with vocal learning, if not proof of it, bats observed to have geographic call variation are good candidates for vocal learning studies. As per a 2019 review, in addition to the bats with direct evidence of vocal learning, 16 bat species (*Hipposideros larvatus, Hipposideros ruber, Pteronotus parnellii, Rhinolophus blasii, Rhinolophus capensis, Rhinolophus clivosus, Rhinolophus cornutus, Rhinolophus damarensis, Rhinolophus euryale, Rhinolophus hipposideros, Rhinolophus mehelyi, Rhinolophus monoceros, Rhinolophus sinicus, Rhinonicteris auranta, Nyctinomops laticaudatus, Thyroptera tricolor*) have been identified to have geographically varying calls (Figure 1). One study found that the divergence in calls between two geographically isolated cryptic species of *Pteronotus parnellii* was unlikely to be explained by genetic drift alone, and could perhaps be best explained by cultural mother-pup transmission, which offers stronger evidence than most geographic variation results (Dávalos et al., 2019). Adult call similarity among the matriline and among relatives offering indirect evidence consistent with but not solely indicative of vocal learning was also observed in *Antrozous pallidus*. Overall, there is indirect evidence for vocal learning in members of eight chiroptera families: Pteropodidae, Hipposideridae and Rhinolophidae from Yinpterochiroptera; and Emballonuridae, Phyllostomidae, Mormoopidae, Thyropteridae and Vespertilionidae from Yangochiroptera (Vernes & Wilkinson, 2020). Further studies are needed to conclusively classify these species on the vocal learning spectrum, but they provide good candidates for future vocal learning research.

Echolocation Primes Bats for Vocal Communication

Although the primary use of echolocation is for spatial orientation, bats also use it for social purposes. The evolution of laryngeal echolocation may have created many of the auditory and neural pathways needed for social communication. Salles et al. hypothesized that the same auditory processing systems are involved in both echolocation and social communication signals, especially at the lower stages of the auditory pathway (2019). Laryngeal echolocators have highly specialized audio processing systems, including large middle ear muscles working with super fast laryngeal muscles to produce sounds ranging between nine and 212 kHz. The only other vertebrates with fast laryngeal signal production, such as songbirds, use this adaptation for vocal communication. Bats probably evolved these muscles primarily for echolocation, but have been shown to use them for both echolocation and social communication. Some bat groups observed to use conspecific echolocation calls as social communication include *Rhinolophus spp., Noctilio albiventris, Eptesicus fuscus, Saccopteryx bilineata, Myotis myotis* and *Myotis capaccinii* (Knörnschild, 2014). DSC bats make fine adjustments to the larynx based on auditory feedback, and so evolved the neural circuitry necessary for fine vocal control. If the fundamental machinery for both echolocation and social communication is the same, bats, especially DSC bats, are well primed for the evolution of vocal learning (Salles et al. 2019). Given this overlap, we might also expect the molecular evolution of echolocation to coincide with the evolution of the genes involved in vocal learning.

Evolution of Echolocation in Bats

Phylogeny of Echolocation in Bats

Echolocation is an unusual sensory adaptation in which an individual analyzes the echoes of an emitted sound to acquire important spatial information. Only a few mammals are known to echolocate, including bats, about 80% of which produce echolocation calls in their larynx for orientation and prey detection. The most recent molecular and comparative phylogenetic evidence suggests that the ancestor of all bats was a small, flying nocturnal insect-eating laryngeal echolocator. The trait was lost in the Old World fruit bats (Pteropodidae), likely in a sensory tradeoff for low-light vision better suited for finding fruit and nectar. However, the physiological and genetic basis for biosonar is retained in some fruit bats, as demonstrated by the tongue-clicking echolocation in one cave-dwelling pteropodid species, *Rousettus aegyptiacus*, and by wing-clicking in at least two other genera. The laryngeal echolocation ability was retained in the superfamily of bats most closely related to Pteropodidae, Rhinolophoidea (rhinolophidae, hipposideridae, craseonycteridae, rhinopomatidae and megadermatidae) (Thiagavel et al., 2018). Together, Pteropodidae and Rhinolophoidea form the clade Yinpterochiroptera, while the three other laryngeal echolocating superfamilies (Emballonuroidea, Noctilionoidea and Vespertilionoidea), comprised of 12 families, are grouped in Yangochiroptera (Figure 2).

Foraging Strategy Drives Evolution of Echolocation

Laryngeal echolocation can be classified into three major types: broadband, narrowband, and constant frequency (CF) with Doppler-shift compensation (DSC). Narrowband calls can be further classified by whether they are multiharmonic or dominated by a single, fundamental harmonic (Jones & Teeling, 2006). Different echolocation call structures evolved multiple times in different bat families, but the ancestor of all bats was believed to use narrowband multiharmonic calls (Thiagavel et al., 2018). While phylogenetic history partially predicts echolocation call structures, pressure to adapt to habitat-specific tasks also played a large role in shaping call type (Russo et al., 2018). Bats hunting in open spaces, such as many species in Vespertilionidae and Molossidae, often use long, narrowband signals in order to detect distant insects. Some bats, such as some of those in Vespertilionidae, switch from narrowband calls while searching for prey to rapid broadband calls while approaching prey or to navigate cluttered areas, since three-dimensional structures are well detected by this call structure. The most sophisticated echolocation type, CF with DSC, evolved multiple times in all of Rhinolophidae and Hipposideridae, some of Noctilionidae, and one lineage in Mormoopidae (*Pteronotus parnellii*) (Figure 2). These species use CF signals with no bandwidth to detect and classify their fast-moving prey with high accuracy in cluttered environments. CF bats use DSC to adjust for the change in frequency induced by their own flight speed. Bats in Rhinolophidae, Hipposideridae and *Pteronotus parnellii* use a slightly more advanced call structure than bats in Noctilionidea, which may emit shorter calls with longer gaps and are not obligate CF emitters (Jones & Teeling, 2006;Thiagavel et al., 2018). As we gain more molecular data, our resolution of bat phylogenetics and locations of echolocation call structure divergence are becoming more clear.

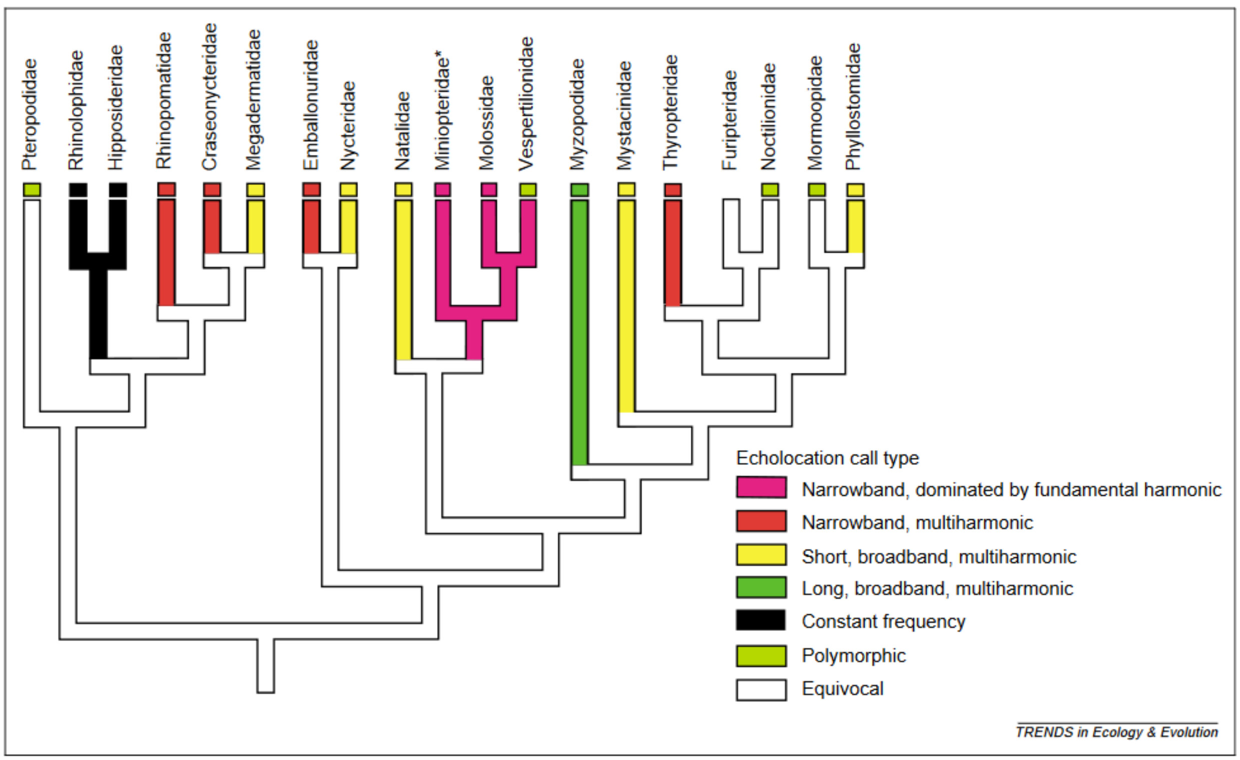

Figure 2. Echolocation types of 19 bat families, from Jones & Teeling, 2006.

Molecular Evolution of Vocal Learning Genes in Bats

Evolutionary pressures shaping vocal learning

The evolutionary processes shaping vocal learning remain poorly understood. Studying the evolution of vocal-learning genes by comparative methods has been limited by the rarity and diversity of lineages with the trait. In the past, evolutionary studies have focused primarily on songbirds, in whom it has evolved in three separate clades. In singing courtship animals, such as songbirds and humpback whales, several hypotheses exist for the benefit of vocal learning, such as the ‘developmental stress hypothesis,’ which suggests that vocal learning occurs during critical periods of developmental stress in male singers, and so quality or complexity of song will indicate fitness to females. Another hypothesis is the ‘environmental adaptation hypothesis,’ in which birds better learn songs able to be heard well in their acoustic environment, and so vocal learning will increase the ability of birds to adapt their songs to their habitat. In humans, the dominant vocal learning hypothesis is the ‘information-sharing’ hypothesis, which suggests that vocal learning is needed for vocabulary expansion so that information can be transmitted to kin, resulting in a fitness benefit. Parrots and dolphins use their learned vocalizations to refer to individuals with contact calls or signature whistles, suggesting that vocal learning evolved to promote social cohesion (Nowicki & Searcy, 2014).

Animals for whom the best evidence of vocal learning is conversion to a group signature, including most bats, likely evolved vocal learning to enhance group recognition (Nowicki & Searcy, 2014). This hypothesis is supported by the existence of many cryptic species of bats who differ primarily by their echolocation frequencies and genetic signals in a possible example of acoustic divergence leading to reproductive isolation and potentially sympatric speciation (Russo et al., 2018). The vocal learning capabilities of many species of bats remain unexamined, but their complex social structure suggests the possibility that some species may have evolved more sophisticated vocal learning techniques such as information-sharing or individual reference. Studying the evolution of vocal learning genes across the bat phylogeny may reveal more about both the phylogenetic history of bats and the pressures shaping vocal learning.

FOXP2 as a Vocal Learning Gene in Bats

FOXP2 is one of the most widely studied vocal learning genes both in humans and in other species. FOXP2 is critical to the development of the neural pathways and sensorimotor coordination required for language processing, speech, and song. In many vertebrates, although not bats, FOXP2 is extremely conserved, with similar expression patterns observed in the brains of humans, monkeys, mice, rats, birds and crocodiles (Fisher & Scharff, 2009). FOXP2 encodes a regulatory protein in the Forkhead-box group of transcription factors. In humans, mutations in the seventh exome of the gene are associated with speech and language impairment. In adult male zebra finches, FOXP2 was found to be significantly downregulated while practicing their song alone, but not while performing for a potential mate (Teramitsu & White, 2006). FOXP2 is expressed more highly in regions of the zebra finch brain associated with vocal learning during the period of development when vocal learning predominantly occurs, while in canaries it is differentially expressed seasonally in association with changes in song stability (Haesler et al., 2004). These results suggest a role for FOXP2 in the auditory pattern learning process in some species.

Unlike in most other mammals, FOXP2 has been found to be extremely diverse in echolocating bats, possibly in association with diverging biosonar systems. A comparative study of FOXP2 across 13 bat species and 26 non-bat species found accelerated evolution and high variability within bats compared to non-bats, particularly within Exons 7 and 17. A follow up survey of these two exons in 42 bat species (from 10 families), 18 cetaceans (from two families) and three outgroups found that variation in both exons was well conserved across families and corresponded with echolocation types and phylogenetic boundaries. Interestingly, Exon 17, which was invariant across non-bat clades, was highly variable in bats and also associated with phylogenetic boundaries. Also, the Exon 7 asparagine-to-serine substitution linked to a human language learning deficit phenotype was found to occur independently in echolocators of Yinpterochiroptera as well as one carnivore outgroup. A convergent amino acid substitution between *Hipposideros* and one population of the cryptic species *Pteronotus parnellii* is particularly interesting given FOXP2s implications in echolocation, since both groups independently evolved DSC (Li et al., 2007). The expression of FOXP2 in various regions of the brain of *Rhinolophus ferrumequinum* was found to be differentially higher in regions associated with echolocation (Yin et al., 2017). The insecticide Imidacloprid was found to decrease FOXP2 expression in the insectivorous bat *Hipposideros armiger terasensis*, and is suspected to disrupt the vocal, auditory, orientation, spatial memory and echolocation systems in bats (Wu et al., 2020).

Other Vocal Learning Gene Candidates in Bats

Several other genes are known to play important roles in the FOXP2 pathway, and thus may be involved in vocal learning. FOXP1, FOXP2 and FOXP4 have functionally redundant subdomains and work in tandem to regulate target gene expression. Knockdowns of FOXP1, FOXP2 and FOXP3 in zebra finches all impact song learning ability with overlapping but distinct phenotypes (Norton et al., 2019). In sexually dimorphic singing regions of mice brains, FOXP1 expression was found to be higher in male mice than female mice, although the same pattern was not found with FOXP2. CNTNAP2 is also involved in the FOXP2 pathway and is associated with language disorder and autism and humans. The expression of CNTNAP2 in two vocal learning bats, *Phyllostomus discolor* and *Rousettus aegyptiacus*, was found to be largely similar to the expression in human brains, except in the cortex of the adult *Phyllostomus discolor*, suggesting that the expression of FOXP2 is not universal in the cortex of vocal- learning mammals (Rodenas-Cuadrado et al., 2018).

Other genes associated with language disorders in humans may also offer good targets for elucidating the evolution of vocal learning in bats. This list includes DCDC2, which is associated with dyslexia, ROBO2, which is associated with language disorder, and SRPX2, which is associated with language disorder, Rolandic seizures and intellectual delay (Mountford & Newbury, 2018). In a study comparing the gene expression profile of humans and zebra finches and their closest non-vocal learning relatives, chickens and macaques, Pfenning et al. found several additional genes likely associated with vocal learning, including ROBO1, SLIT1, GPMA, and GABR3 which were significantly downregulated in vocal learning birds and humans (2014). ROBO1 has previously been associated with dyslexia and speech disorders, and differential expression was found in fetal brain speech areas in both humans and songbirds. The SLIT proteins work in conjunction with the ROBO proteins to aid in the formation of the central nervous system in bilaterians, and the SLIT1 promoter is targeted by FOXP2 (Pfenning et al. 2014). In a scan for accelerated genomic regions specific to vocal learning birds, markers for positive selection were detected in ZEB2 and NEUROD6 in addition to FOXP2 (Cahill et al., 2021). ZEB1 and ZEB2 are transcription factors known to play an important role in vertebrate embryo development, and more recently understood to play an important role in the development of the nervous system. Mutations in ZEB2 in humans result in Mowat–Wilson syndrome (MWS), a neurological condition that cooccurs with abnormal nervous system development and function as well as craniofacial defects (Gheldof et al., 2012; Hegarty et al., 2015). The NEUROD family is associated with the development of the central nervous system, and variants of the NEUROD family in humans are associated with neurological disorders including Alzheimer's disease (Tutukova et al., 2021). Studies of these genes in a broader group of bats may reveal underlying patterns of vocal learning evolution.

Echolocating Gene SLC26A

Given the relationship between vocal learning and echolocation, we may expect genes involved in echolocation to be impacted by similar stressors and involved in similar pathways as vocal production. One key example of this is SLC26A5, also known as *prestin*, a gene imperative to cochlea function and hearing in many species, including bats and humans. The SLC26A5 gene tree is monophylogenetic among echolocating bats, which is in conflict with the species phylogeny, in which echolocation is paraphyletic (Li et al., 2008). An insecticide shown to adversely affect the vocal, auditory, orientation, and spatial memory systems in an echolocating bat showed decreased SLC26A5 expression in the cochlea as well as decreased FOXP2 expression in the brain, demonstrating the link between these sensory pathways (Wu et al., 2020). By studying the evolution of echolocating genes like SLC26A5 in conjunction with vocal learning genes, we can begin to understand how selection pressures may have acted on these interconnected systems and piece together a broader picture of bat evolution.

Conclusion

The capacity for vocal learning has not been studied in the vast majority of bats, yet they offer an intriguing target for studying the molecular evolution of the complex and uncommon trait. Given the highly developed vocal-auditory system required for laryngeal echolocation, most bats already have much of the physiological machinery needed for vocal production learning. They also have highly complex social structures, which is one of the life-history traits associated with selection for vocal communication. Together, these characteristics make bats prime targets for vocal learning research.

Limited research has examined the role of vocal learning genes, such as FOXP2 in bats, but current molecular evidence suggests its role in the bat vocal-auditory pathway. Given the close relationship between echolocation and vocal production learning in the auditory-vocal pathway, we may also expect that selection pressures acting on key echolocation genes, including SLC26A, may be associated with vocal learning genes and phenotypes in bats. Other genes have been implicated in the vocal learning pathways of other organisms but remain unstudied in bats, although they too may help uncover the greater picture of bat adaptation. Future research should examine a wide range of potential vocal learning genes in as many bat species as possible.

The molecular evolution of a complex trait such as vocal production learning can be difficult to study, especially because it seems to have evolved convergently across a wide range of taxonomy. Tying selection pressure to gene function requires perspectives from a range of phylogenetically diverse species with vocal learning traits. Bats, with their diversity of social structure and demonstrated physiological priming for vocal learning, offer an ideal target for vocal learning studies. In the past, vocal learning research on bats has been limited by the narrow number of behavioral studies and amount of genetic data available for bats. Genes involved in vocal learning pathways need to be analyzed in association with behavioral phenotypes to better unravel the molecular basis for vocal learning. As behavioral studies continue to look for and perhaps find additional, direct evidence for vocal learning in bat species, we will be better able to connect genes to functions. Additionally, the advent of the Bat1K initiative, which aims to sequence the genome of all living bat species, will increase the amount of genetic data available and enable both genome-wide association studies and targeted studies of vocal learning genes.

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